

2026 Faculty Surveys

This is our annual faculty program assessment survey for academic year 2026 (2025-2026). The survey is required for all chemical engineering faculty members and is **very important** for our program assessment and future re-accreditation effort in 2026. The survey does three things. First, it documents that you have been made aware of the performance of our cadets on our program's student outcomes. Second, it serves to document your opinions of that performance. Third, it allows us to use your collective knowledge and experience to identify areas where we might need improvement. Thus, the completed surveys are your collective "thumbs up or down" to the various performance indicators we are tracking.

Instructions

- Print your name and date on the top of this page.
- Please review the data in the document "AY25 Program Assessment Data - 16 July 2025." The data pertains to the level of achievement of our cadets for AY2025. Answer the survey questions in "Part I" on page 3 of this document based on your opinions of that data.
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- Finally, there are some open questions in Part III on page 5 where you can comment on the quality of the curriculum, the process itself or any other items you would like us to address.
- Submit the completed document to Dr. Biaglow by COB **Friday 14 August 2026**. Please be prompt or early if possible.
- Direct any questions about the data or survey to Dr. Biaglow.
- Your responses will be consolidated, discussed at our program assessment meetings, and archived in our annual report.

Mission Statement: The mission of the chemical engineering program is to prepare commissioned leaders of character who possess the intellectual capital to leverage new and emerging technologies.

Chemical Engineering Program Objectives: During a career as commissioned officers in the United States Army and beyond, program graduates:

- Demonstrate effective leadership by leveraging chemical engineering expertise and precise technical communication.
- Contribute to the solution of complex problems in a dynamic environment.
- Apply disciplined technical expertise to succeed in advanced study programs.

Chemical Engineering General Program Outcomes (Outcomes 1-7): On completion of the chemical engineering program, our graduates demonstrate an ability to:

- [Student Outcome 1] Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- Communicate effectively with a range of audiences.
- Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
- Function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
- Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- Acquire and apply new knowledge as needed, using appropriate learning strategies.

Chemical Engineering Curriculum Outcomes (Outcome 8): The program provides the graduate with a thorough grounding and working knowledge of the chemical sciences, including:

- Chemistry
- Material and energy balances
- Safety and environmental factors
- Thermodynamics of physical and chemical equilibria
- Heat, mass, and momentum transfer
- Chemical reaction engineering
- Continuous and staged separation operations
- Process dynamics and control
- Modern experimental and computing techniques
- Process design

Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

The cadets in the program are able to:	Strongly Disagree		Neutral		Strongly Agree
• Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	☐	☐	☐	☐	☒
• Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	☐	☐	☐	☐	☒
• Communicate effectively with a range of audiences.	☐	☐	☐	☐	☒
• Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	☐	☐	☐	☐	☒
• Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	☐	☐	☐	☐	☒
• Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☐	☒
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
The student outcomes are consistent with the program mission and objectives.	☐	☐	☐	☐	☒
The program has a process for periodically assessing the achievement of its student outcomes.	☐	☐	☐	☐	☒
The survey methods used by the program are effective.	☐	☐	☐	☐	☒
The cadets in the program are aware of the program objectives.	☐	☐	☐	☐	☒
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☐	☒
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Name: Biaglow

Date: 7/15/26

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

I would like to see an engineering measurements course happen before the senior lab to focus on sensors, measuring and recording signals, to better prepare the cadets for the senior lab.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes.

Please add any additional comments that you would like to make below.

None.

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• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

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The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
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The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☐	☒
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The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Assessment data: consider supplementing engineering mathematics with differential equations as able throughout chemE curriculum (e.g., transport and kinetics)

Personal opinion: consider introducing electrochemical engineering (reinforces conservation, kinetics, thermodynamics, and transport phenomena in a field where chemEs are currently in high demand) to the curriculum as courseload permits. This would also support material science learning objectives.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes! To continually improve assessment, if possible, we could include more data to mirror the topical outcome evaluations with the end-of-semester surveys. If cadets feel they're not gaining value from certain courses (e.g., EE301, MC300), perhaps we should question if that's due to the content (high scores, low interest) or aptitude/investment (low scores, low interest) and we can supplement shortcomings internally as able.

Please add any additional comments that you would like to make below.

Are EE301 and MC300 ABET-mandated courses? If not, can we choose more chemE-specific courses to transition cadets to chemE curriculum earlier?

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• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
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The cadets in the program are aware of the program objectives.	☐	☐	☐	☐	☒
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☐	☒
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Course curriculum is solid; all core topics in chemical engineering are thoroughly addressed and the year-to-year performance on the FEE reflects broad coverage.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Surveys are complete and effective.

Please add any additional comments that you would like to make below.

No additional comments. Great program.

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Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☒	☐
Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☒	☐

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I think the major curriculum is sufficient within the USMA academic requirement constraints. Most of my advisees are interested in at least 1 BioE course, but restrictions on enrollment and the number of sections that can be offered make it difficult to get everyone into the classes.

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• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☒	☐
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
The student outcomes are consistent with the program mission and objectives.	☐	☐	☐	☐	☒
The program has a process for periodically assessing the achievement of its student outcomes.	☐	☐	☐	☐	☒
The survey methods used by the program are effective.	☐	☐	☐	☒	☐
The cadets in the program are aware of the program objectives.	☐	☐	☒	☐	☐
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☒	☐	☐
The cadets are satisfied with the courses in the program.	☐	☐	☐	☒	☐
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

I think we still need to take time to develop a fully vetted coding complement to the curriculum.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

These are the right questions to ask. I cannot think of any further questions.

Please add any additional comments that you would like to make below.

Feedback for "Communicate effectively with a range of audiences." We should try to figure out ways to have the Cadets brief individuals that do not have implicit Chemical Engineering knowledge. This will also be a necessary skill in the Army where they need to communicate their Unit's needs to other units or even branches of service.

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- [Student Outcome 1] Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- Communicate effectively with a range of audiences.
- Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
- Function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
- Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- Acquire and apply new knowledge as needed, using appropriate learning strategies.

Chemical Engineering Curriculum Outcomes (Outcome 8): The program provides the graduate with a thorough grounding and working knowledge of the chemical sciences, including:

- Chemistry
- Material and energy balances
- Safety and environmental factors
- Thermodynamics of physical and chemical equilibria
- Heat, mass, and momentum transfer
- Chemical reaction engineering
- Continuous and staged separation operations
- Process dynamics and control
- Modern experimental and computing techniques
- Process design

Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

The cadets in the program are able to:	Strongly Disagree		Neutral		Strongly Agree
• Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	☐	☐	☐	☐	☒
• Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	☐	☐	☐	☒	☐
• Communicate effectively with a range of audiences.	☐	☐	☐	☒	☐
• Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	☐	☐	☐	☒	☐
• Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	☐	☐	☐	☐	☒
• Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☐	☒
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
The student outcomes are consistent with the program mission and objectives.	☐	☐	☐	☐	☒
The program has a process for periodically assessing the achievement of its student outcomes.	☐	☐	☐	☐	☒
The survey methods used by the program are effective.	☐	☐	☐	☒	☐
The cadets in the program are aware of the program objectives.	☐	☐	☐	☒	☐
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☒	☐
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

An elective on membrane separation technology would be a great addition. From what I recall in CH363, membranes are taught but not intensively. Membranes are a great way to take advantage of chemical properties to separate desired products, and we could easily fabricate/fix the membrane in the sub-basement to show cadets how to measure pressure differentials to determine the effectiveness of membranes.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Nothing that I can think of.

Please add any additional comments that you would like to make below.

One of the most common issues is that cadets seem to forget content after completing a course. Instructors often remark that what is taught is not retained. I wonder if the program could implement a portfolio of work that cadets maintain on their own, and could contribute to them earning their degree with honors. I envision the course directors could have an option assignment, issued after the TEE, where cadets have the opportunity to practice what they learned in that course outside of the classroom. Cadets who complete all optional assignments could graduate with honors. The benefit would be observed in CH400 and CH459, where they have to use all ChemE concepts.

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• Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	☐	☐	☐	☐	☒
• Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☒	☐
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☐	☒
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
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The survey methods used by the program are effective.	☐	☐	☐	☐	☒
The cadets in the program are aware of the program objectives.	☐	☐	☐	☐	☒
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☒	☐
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Name: COL Corey James

Date: 8/5/26

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Yes, we are teaching the right courses. If I were to add a course, it would be another semester of design.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

N/A

Please add any additional comments that you would like to make below.

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Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

The cadets in the program are able to:	Strongly Disagree		Neutral		Strongly Agree
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• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
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The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☐	☐	☒
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Yes, the program is teaching the necessary foundational chemical engineering courses. There is a need to have a design course II. Recommend adding 1.5 credit hours to CH400 and have lessons after the FE focused on the CH402 capstone. This will be effective once the program removes the requirement for 3x engineering electives and makes 2x engineering electives required.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes, No comment.

Please add any additional comments that you would like to make below.

The honors requirement for the major should be changed following the ABET visit. The requirement should be 3.5 major course GPA and passing the FE Exam.

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Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Name: Plante, Luke T.

Date: 7/16/26

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
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The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
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Name: Plante, Luke T.

Date: 7/16/26

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Consider increasing thermal radiation emphasis in the CH485 Heat and Mass Transfer.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes - the questions seem appropriate for both this year and year.

Please add any additional comments that you would like to make below.

Consider adding more chemical engineering courses to first semester of sophomore year.

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- [Student Outcome 1] Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- Communicate effectively with a range of audiences.
- Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
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- Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- Acquire and apply new knowledge as needed, using appropriate learning strategies.

Chemical Engineering Curriculum Outcomes (Outcome 8): The program provides the graduate with a thorough grounding and working knowledge of the chemical sciences, including:

- Chemistry
- Material and energy balances
- Safety and environmental factors
- Thermodynamics of physical and chemical equilibria
- Heat, mass, and momentum transfer
- Chemical reaction engineering
- Continuous and staged separation operations
- Process dynamics and control
- Modern experimental and computing techniques
- Process design

Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

The cadets in the program are able to:	Strongly Disagree		Neutral		Strongly Agree
• Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	☐	☐	☐	☐	☒
• Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	☐	☐	☐	☒	☐
• Communicate effectively with a range of audiences.	☐	☐	☐	☐	☒
• Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	☐	☐	☐	☒	☐
• Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	☐	☐	☐	☐	☒
• Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☒	☐
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☐	☒
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
The student outcomes are consistent with the program mission and objectives.	☐	☐	☐	☐	☒
The program has a process for periodically assessing the achievement of its student outcomes.	☐	☐	☐	☐	☒
The survey methods used by the program are effective.	☐	☐	☐	☐	☒
The cadets in the program are aware of the program objectives.	☐	☐	☒	☐	☐
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☒	☐
The cadets are satisfied with the courses in the program.	☐	☐	☐	☐	☒
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

No change to curriculum.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

No change to the faculty survey.

Please add any additional comments that you would like to make below.

The ABET Advisory Board visit highlighted the need to address where and how the Chem E Field is changing with time. Our program would benefit from establishing a set of focus areas that can be integrated across lesson content to provide cadets early exposure and build future competencies. Ex. Extracting python code from publications for optimization, Address Energy Security through Biomass, Materials, etc. Theme can also focus on certain FE topical outcomes to build/maintain performance above national average.

2026 Faculty Surveys

This is our annual faculty program assessment survey for academic year 2026 (2025-2026). The survey is required for all chemical engineering faculty members and is **very important** for our program assessment and future re-accreditation effort. The survey does three things. First, it documents that you have been made aware of the performance of our cadets on our program's student outcomes. Second, it serves to document your opinions of that performance. Third, it allows us to use your collective knowledge and experience to identify areas where we might need improvement. Thus, the completed surveys are your collective "thumbs up or down" to the various performance indicators we are tracking.

Instructions

- Print your name and date on the top of this page.
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- The survey also asks additional questions pertaining to the program objectives in "Part II" on page 4. For this part of the survey, we are interested in your opinion of the relevance of the objectives and their consistency with the Academy mission and needs of the Army.
- Finally, there are some open questions in Part III on page 5 where you can comment on the quality of the curriculum, the process itself or any other items you would like us to address.
- Submit the completed document to Dr. Biaglow by COB **Friday 14 August 2026**. Please be prompt or early if possible.
- Direct any questions about the data or survey to Dr. Biaglow or Dr. Nagelli.
- Your responses will be consolidated, discussed at our program assessment meetings, and archived in our annual report.

Mission Statement: The mission of the chemical engineering program is to prepare commissioned leaders of character who possess the intellectual capital to leverage new and emerging technologies.

Chemical Engineering Program Objectives: During a career as commissioned officers in the United States Army and beyond, program graduates:

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Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

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• Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	☐	☐	☐	☐	☒
• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☐	☒
The program curriculum supports the program objectives.	☐	☐	☐	☐	☒
The student outcomes are consistent with the program mission and objectives.	☐	☐	☐	☐	☒
The program has a process for periodically assessing the achievement of its student outcomes.	☐	☐	☐	☐	☒
The survey methods used by the program are effective.	☐	☐	☐	☒	☐
The cadets in the program are aware of the program objectives.	☐	☐	☐	☒	☐
The cadets are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒
The cadets are satisfied with the courses in the program.	☐	☐	☒	☐	☐
The faculty are aware of the program objectives.	☐	☐	☐	☐	☒
The faculty are given an opportunity to provide their opinion about the program objectives.	☐	☐	☐	☐	☒

Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Our program is grounded in the three pillars of Chemical Engineering, and I think we are on the right track. An applications-focused survey course could be useful, but I do not think it is required to meet our end state.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes, questions assess faculty awareness of student outcomes, and our opinion of those outcomes.

For additions, have we considered the ability to select not-observed or N/A for appropriate fields?

Please add any additional comments that you would like to make below.

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Part I. Student Outcomes. Review the data and then mark the box in the column that most closely represents your opinion.

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• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

Part II. Program Objectives. Mark the box that most closely represents your opinion.

	Strongly Disagree		Neutral		Strongly Agree
The program objectives are consistent with the USMA mission.	☐	☐	☐	☐	☒
The program objectives are consistent with the needs of the Army.	☐	☐	☐	☒	☐
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Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Overall, I think we're teaching the right courses and material given the many constraints on an engineering program at USMA.

I'll reiterate my previous recommendation: I think we should remove CH102 as a pre-requisite for CH362 to move CH362 up to the first semester of yearling year, which would enable the program to spread across 6 semesters instead of only 5. We should also facilitate open-ended inquiry by allowing CH489 to count as an elective, if there is sufficient engineering content to the research project.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

You could add a few questions along these lines:

- 1) How likely are you to recommend the Chemical Engineering program to a cadet who is interested in STEM?
- 2) How likely are you to recommend teaching in CBSE to a peer who is qualified to teach in multiple departments?

Please add any additional comments that you would like to make below.

I think the program should take a more deliberate approach to junior faculty development, including exposure for all junior faculty to teaching (or assisting in teaching) the higher level courses.

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• Acquire and apply new knowledge as needed, using appropriate learning strategies.	☐	☐	☐	☐	☒
• Have attained a thorough grounding in and working knowledge of the chemical engineering curriculum.	☐	☐	☐	☐	☒

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	Strongly Disagree		Neutral		Strongly Agree
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Part III. Open Questions.

Are we teaching the right classes? Based on the assessment data or on your personal opinion, is there a course or content area that the program should add to the curriculum?

Yes, I believe we are teaching the right classes.

Are we asking the right questions? Do you have any suggestions to improve the faculty survey for next year?

Yes, I believe we are asking the right questions.

Please add any additional comments that you would like to make below.

As AI tools become increasingly prominent, integrating structured AI instruction throughout the curriculum would be highly beneficial, moving cadets away from entering vague prompts and toward effective, intentional usage.